INCH-POUND

MIL-DTL-40061C
1 June 2026
SUPERSEDING
MIL-P-40061B
31 December 1970

MILITARY SPECIFICATION

PANEL MARKER, AERIAL LIAISON, VS-17/GVX

This specification is approved for use by all
Department and Agencies of the Department of Defense

1: SCOPE

1.1: Scope This specification covers one type and size of signal panel which is fabricated from laminated fluorescent dyed nylon cloth with an integral pocket for protection of the panel in storage.

2: APPLICABLE PUBLICATIONS

2.1: General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether they are listed.

2.2: Government documents.

2.2.1: Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense and supplement thereto, cited in the solicitation (see 6.2).

FEDERAL SPECIFICATIONS

A-A-59963	- Thread, Polyester
UU-B-36	- Bag, Paper (grocers)
A-A-203	- Paper, Kraft, Wrapping

Comments, suggestions, or questions on this document should be addressed to The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or email to usarmy.belvoir.hqda.mil.tioh-webmaster@army.mil. You may verify the currency of this specification using the ASSIST Online database at https://assist.dla.mil.

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MILITARY SPECIFICATIONS

MIL-DTL-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-DTL-10884 - Fastener, Snap
PIA-C-5040 - Cord, Fibrous, Nylon
MIL-DTL-32075 - Label for Clothing, Equipage, and Tentage (general use)

FEDERAL STANDARDS

FED-STD-191 - Textile, Test Method

MILITARY STANDARD

MIL-STD-129 - Military Marking for Shipment and Storage
MIL-STD-810 - Environmental Engineering Considerations and Tests

(Copies are available online at https://assist.dla.mil)

2.2.2: Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

THE INSTITUTE OF HERALDRY DRAWINGS

5-1-321 - Panel Marker, Aerial Liaison, VS-17/GVX

Assembly (illustrated as Figure 1)

(Reduced size copies of the above drawings are included in the specification to illustrate the different types, styles, classes, and construction of buttons. Full size versions of TIOH numbered drawings only can be obtained from the procuring activity or as directed by the contracting officer.)

2.3: Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are issues of the documents cited in the solicitation (see 6.2).

ASTM INTERNATIONAL

ASTM-D6251 - Boxes, Wood, Cleated-Plywood
ASTM-D6880 - Boxes, Wood Nailed and Lock-Corner
ASTM-D1974/D1974M - Fiberboard Boxes, Methods of Closing, Sealing and Reinforcing, Standard Practice for
ASTM-D5118/D5118M - Boxes, Fiberboard Shipping, Fabrication of
ASTM-D3776 - Fabric, Mass per Unit Area
ASTM-D751 - Fabric, Coated

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ASTM-D2261 - Fabric, By the Tongue (single rip) procedure (constant-rate-of-extension tensile testing machine), Tearing strength of
ASTM-D3775 - Standard Test Method for End (wrap) and Pick (filling) Count of woven Fabric
ASTM-D6193 - Stitches, Seams, and Stitching's

(Copies of these documents are available online at http://www.astm.org)

2.4: Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3: REQUIREMENTS

3.1: Standard Sample Unless otherwise specified in the contract or purchase order (see 6.2), the contracting officer will furnish a standard sample of each laminated cloth (red-orange and olive drab-orange) which shall be used for matching color, luster, brightness, purity of tone and colorfastness only.

3.2: First Article When specified (see 6.2), before production commences, a sample of the finished button shall be submitted or made available to the contracting officer or authorized representative for inspection and tests as specified in 4.2. The approval of the first article authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with the provision of this specification. The first article shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.3: Materials. Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1: Cloth, nylon. The nylon cloth used for the panel marker shall be a five-harness warp sateen, counter 2, which has the following characteristics:

a. A minimum of 160 ends per inch in the warp and a minimum of 80 picks per inch in the filling
b. At least 70 denier yarn shall be used in the warp and at least 100 denier yarn shall be used in the filling.
c. Maximum weight of 3 ounce per square yard.
d. The cloth shall be heat treated.
e. The cloth shall be either pieced dyed or yarn dyed.

3.3.1.1: Color. The nylon cloth used in fabrication of the panels shall be of the following colors: Fluorescent red, fluorescent orange and olive drab No. 7,as required. The fluorescent red shall match US Army Shade 229 and the fluorescent orange shall match US Army Shade 230.

3.3.2: Tie tape. The tape for the tie shall be dyed to approximate olive-green Shade No. 106. The tape shall conform to type III of MIL-DTL-5038, except only the following characteristics shall apply: Material identification, color, width, colorfastness to light, ultimate elongation and breaking strength. Breaking strength shall be not less than 100 pounds.

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3.3.3: Thread Polyester. Thread for stitching shall be natural colored nylon thread, conforming to type I or type II, Class1, size B, 2 or 3 ply of A-A-59963, except that only the following characteristic shall apply: Material identification, breaking strength and elongation.

3.3.4: Cord nylon. The cord for the tie cords shall be nylon cord dyed to approximate olive-green Shade No. 106 and shall conform to type II of PIA-C-5040, except that only the following characteristics shall apply: Material identification, number of carriers and ends, color, colorfastness to light and a minimum breaking strength of 100 pounds.

3.3,5: Snap Fasteners. The snap fasteners shall consist of a socket and stud eyelet combination conforming to style 2, finish 2 of MIL-DTL-10884.

3.3.6: Adhesive. Any resin type adhesive may be used in the lamination of the nylon fabric, provided when tested as specified as 4.3.1, the minimum adhesive strength of the lamination shall be 4-1/2 pounds, and that adhesive does not show signs of cracking, flaking or peeling because of the test.

3.4: Construction. The panel marker shall be constructed in accordance with TIOH drawing 5-1-321 and the requirements specified herein.

3.4.1: Stitching. All stitching shall be machine stitching conforming to ASTM-D6193, for all stitching, except for tackling, stich type 301 with 6 to 8 stitches per inch shall be used. Bar tacks shall be a minimum of 5/8 inch wide with not less than 28 stitches. Thread tension shall be maintained so that no loose stitching shall occur. All thread ends shall be trimmed. When thread breaks occur in the type 301 stitching, the break shall be backstitched not less than one inch. Except where ends are turned under in a hem or held down by other stitching the ends of all stitching shall be backstitched not less than ½ inch.

3.4.2: Seating of snap fastener. Snap fasteners shall be securely clinched to the panel marker. If a hole is pouched through the material to aid in the placement of the fastener, the diameter of the hole shall be sufficiently small to require the stud portion of the stud eyelet combination to be forced through the hole. Care should be taken when clinching the fastener to the material to ensure the fabric is not cut or damaged.

3.4.3: Tape and cord. The tape and cord shall be attached to the panel marker as shown on TIOH drawing 5-1-321. The ends of both the tape and cord shall be heat sealed to prevent raveling.

3.4.4: Laminated cloth. Two pieces of nylon cloth specified in 3.3.1 shall be laminated back-to-back using adhesive specified in 3.3.6 to form two plies in the following two-color combinations:

1. Fluorescent red on one side and fluorescent orange on the other side.
2. Fluorescent orange on one side and olive drab No. 7 on the other side.

3.4.4.1: Characteristics of laminated cloth. The laminated cloth shall be as specified herein.

3.4.4.1.1: Weight. When tested as specified in 4.3.1, the weight of one square yard of laminated cloth shall not exceed 9-1/2 ounces.

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3.4.4.1.2: Flexibility. When tested as specified in 4.4.1 the laminated cloth shall be pliable and shall not crack, flake or peel.

3.4.4.1.3: Breaking strength. When tested as specified in 4.3.1, the laminated cloth shall have a breaking strength of not less than 195 pounds in both the warp and the filling direction.

3.4.4.1.4: Tear resistance. When tested as specified in 4.3.1, the laminated cloth shall have a tear resistance of not less than 9 pounds in both the warp and filling direction.

3.4.4.1.5: Temperature and humidity. When tested as specified in 4.4.2, the nylon cloth shall not crack, flake, discolor or delaminate.

3.4.4.1.6: Streaks and spots. The laminated cloth shall be acceptable when in any yard length, a total area of not more than 4 square inches is streaked or spotted. There shall be no larger density of spots or streaks in any single yard.

3.4.4.1.7: Evenness of lamination. There shall be no lumpy or thick areas in the laminated cloth.

3.4.4.2: Color. Color of the laminated cloth shall be as specified in 3.3.1.1 and 3.4.4.

3.4.4.2.1: Visual color matching. The finished laminated nylon cloth shall match the standard sample (see 3.1 and 6.3) and shall be equal or superior to the standard sample with respect to luster, brightness and purity of color tone when viewed under North Sky daylight, or artificial light having a color temperature of 7500 degrees Kelvin. The samples shall be oriented vertically with the warp direction. The observer shall view the samples at a minimum distance of 6 feet. Viewing shall be made perpendicular to the surface of the sample.

3.4.4.2.2: Colorfastness. When tested as specified in 4.4.3, the laminated nylon cloth shall show color characteristics and fastness to accelerate weathering equal to or better than the standard sample.

3.5: Marking. Each panel shall be marked as specified on TIOH drawing 5-1-321. The markings shall conform to type IV, class 9 of MIL-DTL-32075. The markings show fastness to accelerated weathering for 40 standard hours, equal to or better than the standard sample when tested in accordance with MIL-DTL-32075. When no standard sample has been established for fastness to printing, the markings show good fastness to accelerate weathering when tested in accordance with MIL-DTL-32075.

3.6: Workmanship The finished panel marker shall be clean, well-made and shall meet the acceptable quality levels established by this specification.

4: VERIFICATIONS

4.1: Inspection responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use their own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2: First Article inspection. When a first article is required, the first article shall be inspected and tested for all requirements applicable to the end product testing and examination.

4.3: Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be in accordance with MIL-STD-105, except where otherwise indicated.

4.3.1: Testing of components and materials. In addition to the quality assurance provisions of the referenced specifications and standards, inspection shall be performed on components and materials listed in table I for test characteristic shown. The sample unit for all tests shall be 3 yards, full width. The requirements are applicable to the individual unit, and the lot size can be expressed in yards. For examinations applicable to table I, the acceptance 7 number shall be 0 and the sample size for the designated lot size shall be as follows:

LOT SIZE	SAMPLE SIZE
800 or less	2
801 to 22,000	3
22001 and over	5

TABLE I Testing of Components and Materials

Component	Characteristic	Rqm't para	Test method	Determ per s/u	Results reported as	S/U (inches)
Nylon cloth	Weave	3.3.1	Visual	1	Pass or fail	1 pc (10x10)
	Yarn per inch	3.3.1	ASTM-D3775	5	Yarns in warp filling	5 pc (3x3)
	Weight	3.3.1	ASTM-D3776	5	Nearest 1 oz per sq yd	5 pc (2x2)
Laminated nylon cloth	Colorfastness to accelerated weathering	3.4.4.2.2	AATCC-169	4	Pass or fail (equal to std sample)	4 pc (2x2)
	Color	3.4.4.2		1	Pass or fail	1 pc (12x12)
	Adhesion	3.3.6		5	Nearest .1 lb per 2 inch	
	Breaking strength	3.4.4.1.3	AATCC-EP9	5 ea. direction	Nearest 1.0 lb	5 pc (2x6)
	Tear resistant	3.4.4.1.4	AATCC-EP9	5 ea. direction	Nearest 1.0 lb	10 pc (4x6)
	Weight	3.4.4.1.1			Nearest .1 oz per sq yd	10 pc (3x8)
	Flexibility	3.4.4.1.2	ASTM-D5034	5	Pass or fail	5 pc (2x2)
	Temperature for humidity:	3.4.4.1.5	ASTM-D2261			
	-Low temperature			4	Pass or fail	4 pc (7x7)
	-High temperature high and humidity high		ASTM-D2261	3	Pass or fail	3 pc (8x8)

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	-High temperature and low humidity -Extreme temperature		4.4.1 5874* 4.4.2.2 4.4.2.3 4.4.2.4	4 4 4	Pass or fail Pass or fail	4 pc (4x6) 4 pc (4x6) 4 pc (4x6)

* Methods are specified in FED-STD-191

4.3.1.1: Certification. The contractor shall certify that the cloth used in lamination has been heat treated in accordance with standard commercial practice and that the denier is a minimum of 70 (warp) and 100 (filling).

4.3.2: Inspection of the end items.

4.3.2.1: Visual examination. Defects found during the visual examination shall be classified in accordance with 4.3.2.1.1 and 4.3.2.1.2. The unit of product for these examinations shall be one panel marker. The inspection levels and the acceptable quality levels (AQL's) shall be as specified in 4.3.2.1.3. The lot size shall be expressed in units of one panel marker.

Table II Classification of Defects

Examine	Defect	Classification Major	Minor
Materials Laminated cloths	Spots or streaks exceeding a total of 4 sq. ins. in total area in any one yard of length.....	X	-
	Lumps or thick areas in laminating cloth.....	-	X
	Any component other than specified.....	X	-
	Hole, cut, tear, mend, patch, needle chew or otherwise impaired affecting serviceability.....	X	-
Nylon cords and tape	One or more missing.....	X	-
	Not specified type, size or color.....	-	X
	Cut, hole or tear affecting serviceability.....	X	-
	Ends not heat-sealed to prevent raveling.....	-	X
Snap fasteners	Not securely attached in specified manner.....	X	-
	Not specified type.....	X	-
	More than one missing.....	X	-
	One missing.....	-	X
	Any sharp burr, sliver or edge that may damage panel or cause injury in handling.....	X	-
	Improperly assemble; improperly set, not securely clinched, will not engage socket or otherwise defective, affecting serviceability.....	X	-
	Grommet clinched excessively tight cutting fabric.....	X	-
Reinforcement patches	Missing or not securely attached as specified.....	X	-

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Pocket	Missing……..	X	-
	Not specified color……..	-	X
	Pocket opening corner not reinforced with bar tacking…….	-	X
	Outer edge of pocket stitched to end of panel with single row of stitching or two rows spaced less than 1-3/16 inch or more than 1-5/16 apart……..	-	X
	Outer edges of pocket set uneven with edges of panel by more than 1/16 inch…….	-	X
Seams and stitching's	Other than specified……..…	-	X
	Open more than one inch……	X	-
	Open more than ½ inch, but not more than one inch……..…	-	X
	(A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuously skipped stitches or run-offs occur. On double stitched seams, a seam is considered open when either one or both sides of the seam are open)		
	Run-offs (see open seams)…….	-	X
	Seams badly twisted, pleated or puckered……..…	-	X
	Tension too loose or too tight……..…	-	X
	Three or more stitches per inch less than the minimum specified……..…	X	-
	Two stitches less than minimum as specified…….	-	X
	Two or more stitches in excess of the maximum specified…	-	X
	(Variation in the number of stitches per inch caused by the operator's speeding up the machine and pulling the fabric to sew over heavy seams or heavy places, or in turning corners, shall be classified as follows. a. Within the minor defect classification- no defect b. Within the major defect classification- minor defect.)		
	Tread breaks in type 301 stitching did not backstitch at least 1 inch at each break……	-	X
	Type 301 stitching not backstitched at the ends, except where ends are turned under in a hem or held down by other stitching…….	-	X
	Stitching margins irregular, i.e., unevenly gauged, clearly noticeable…….	-	X
	Bar tack missing, loose or misplaced……..	-	X
	Ends not trimmed……..…	-	X
	Missing……..….	X	-
	Misplaced, misspelled, smeared, type of character not specified or wrong color……	-	X
Marking (Contractor's identification; instructions for care of panel one of each required on inside of pocket, 00 side)	Missing…….	X	-
	Misplaced, misspelled, smeared, type of character not as specified or wrong color…….	-	X

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4.3.2.1.2: Dimensional examination. Any dimension that is not within the specified tolerance shall be classified as a defect.

4.3.2.1.3: Inspection levels and acceptable quality levels (AQL's). The inspection levels and AQL's expressed in defects per hundred units shall be as follows:

	Inspection Level	AQL's Major	AQL's Total
For defects applicable to 4.3.3.1.1	II	4.0	10.0
For defects applicable to 4.3.3.1.3	S-2	(one class)	4.0

4.3.2.1.4: Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Shipping containers fully prepared for delivery shall be examined for the defects of closure listed below. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level will be S-2, and the acceptable quality level shall be 2.5.

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible or improper size, location, sequence or method of application.
Materials	Any component missing, damaged or affecting Serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flags, inadequate stapling.
Content	Number per container is more or less than required.

4.4: Tests.

4.4.1: Flexing test. A specimen of the laminated nylon cloth shall be subject to 1000 flexing cycles at -40 degrees F. and then another 1000 flexing cycles at +150 degrees F. A total of four specimens shall be tested as above, two with the warp placed in the vertical position (opposite faces shall be exposed) and two with filling direction placed in vertical position (opposite faces exposed). The specimens shall be 7 inches by 7 inches, capable of being formed into a cylinder 7 inches long and 2 inches in diameter with a 5/8 inch overlap. The apparatus used shall be one which enables the cylinder described to be flexed through a vertical reciprocating motion a distance of 4 ½ inches. The machine should be capable of maintaining a speed of 150 to 175 cycles per minute at specified test temperatures. In the unflexed and flexed positions, the mounting cylinders shall be 5 inches and ½ inch apart respectively, to produce a 4 ½ inch flexing stroke. The specimen shall be mounted on the specimen holder by means of a ring or hose clamp and the apparatus placed in the test condition specified for one hour. The cloth shall then be examined to determine compliance with 3.4.4.1.2.

4.4.2: Temperature and humidity test.

4.4.2.1: Low temperature test. The low temperature test shall be specified in method 5874 of FED-STD-191. The cloth shall then be examined to determine compliance with 3.4.4.1.5

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4.4.2.2: High temperature and high humidity. Four specimens of the laminated nylon cloth, each 4 inches by 6 inches, shall be subject to a temperature of +150 degrees F. The cloth shall then be examined to determine compliance with 3.4.4.1.5.

4.4.2.3: High temperature and low humidity. Four specimens of the laminated nylon cloth, each 4 inches by 6 inches, shall be subject to a temperature of +150 degrees F. at a relative humidity of 20 ± 5 percent for a period of 14 days. Immediately upon removal from the test chamber, the specimens shall be tested as follows: Two specimens of the laminated cloth shall be bent through 180 degrees around a 3/16 inch diameter steel rod, in the direction of the warp with the orange side of one and red or olive drab side of the other exposed. These 180 degrees bend shall be repeated in the direction of the filling with the remaining two specimens. The cloth shall then be examined to determine compliance with 3.4.4.1.5

4.4.2.4: Extreme temperature. Four specimens of the laminated nylon cloth each 4 inches by 6 inches shall be subjected to steps 2A, 6A and 10 of the temperature cycle shown in MIL-STD-810, except step 6A shall be -60 degrees F., +5 degrees, -0 degrees F. The cloth shall then be examined to determine compliance with 3.4.4.1.5.

4.4.3: Accelerated weathering test. Four specimens on each side of the laminated nylon cloth used in the panels with a piece of the standard sample shall be exposed in accordance with method 5671 of FED-STD-191 with a 40-hour exposure cycle. The apparatus shall be as specified in method 5804 except that the filters shall be removed when testing. The cloth shall then be examined to determine compliance with 3.4.4.2.2.

5: PACKAGING

5.1: Packaging. For acquisition purposes, the packaging requirements shall be level A or C as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the person responsible for packaging activity.

6: NOTES.

6.1: Intended use. Panel marker by this specification is intended to be used by ground forces to affect recognition by air support units.

6.2: Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When standard samples will not be furnished (See 3.1).
(c) When first articles are required (See 3.2).
(d) Selection of applicable levels of packing (See 5.1).

6.3: Standard sample. Standard sample as specified in 3.1 may be obtained from procurement office, issuing the invitation for bid or request for proposal.

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6.4 Subject term (key word) listing.
Acquisition
Acetone
Accelerated
Extreme
Workmanship

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

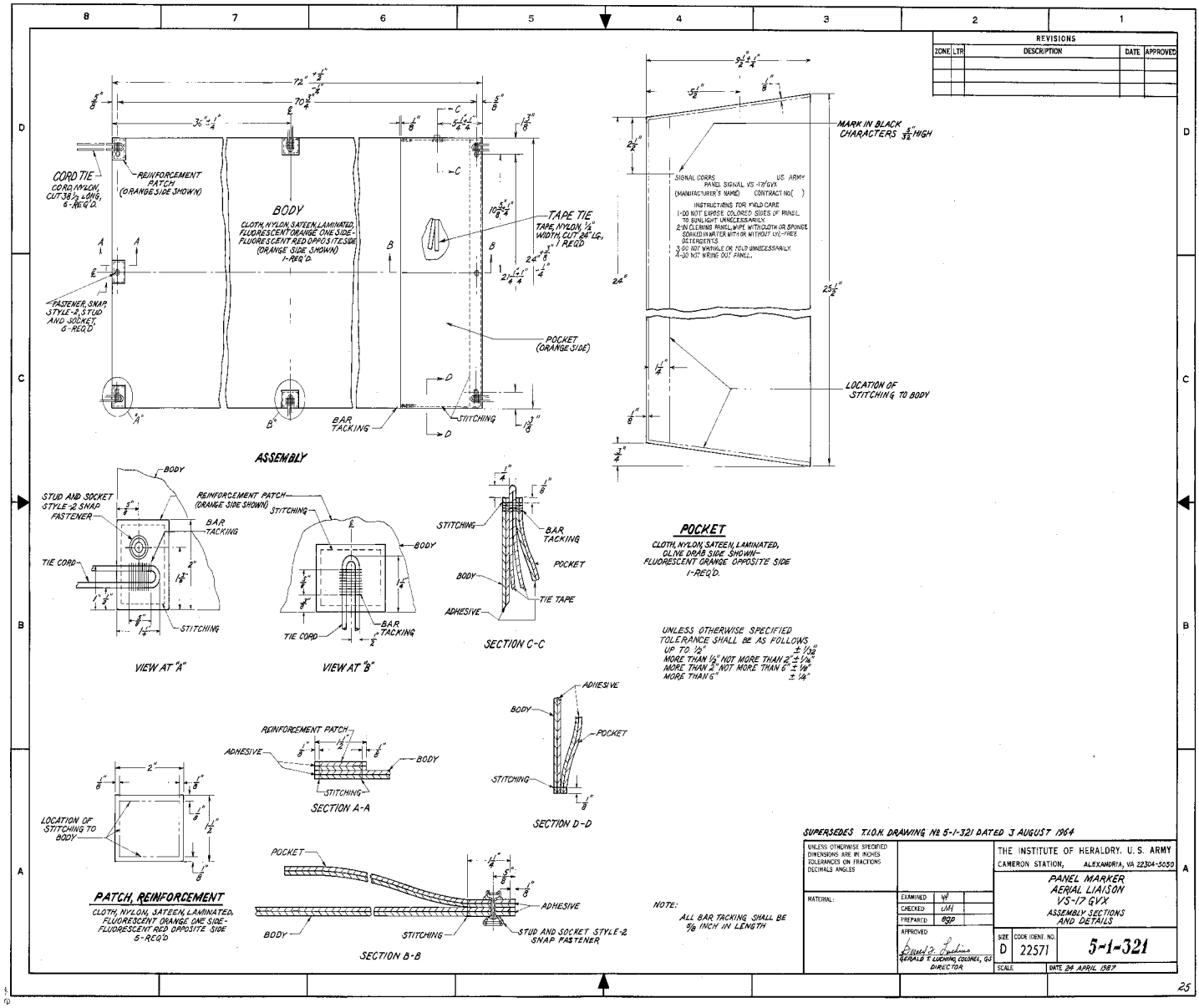


Figure 1. TIOH Drawing No. 5-1-321, Panel Marker

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Army - IH
Navy - NU
Air force - 11

Preparing activity:
Army - IH

Review activities:
Navy - MC, CG1
DLA - CT

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